

Calculus and Linear Algebra

Domain of a Function

Lecture 11

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Lecture outline

- ▶ Natural domain and interval notation.
- ▶ Rational functions and forbidden denominator values.
- ▶ Even and odd roots.
- ▶ Logarithms and trigonometric restrictions.
- ▶ Combining several restrictions.
- ▶ Nested functions and compositions.
- ▶ Reading the domain from a graph.
- ▶ A systematic algorithm and progressively harder exercises.

Main goal

Given a formula, determine exactly **where the function exists as a real-valued function**.

Recall from Lecture 10

A real-valued function is written as

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}.$$

The set D is its **domain**: the set of admissible inputs. For a formula such as

$$f(x) = \frac{1}{x - 2},$$

not every real number can be used as an input.

Question of the day

For which values of x is the formula mathematically meaningful?

Natural domain

The **natural domain** of a real-valued function is the largest subset of $\mathbb{R}$ on which its formula is well defined.

$$D_f = \{x \in \mathbb{R} : \text{the expression } f(x) \text{ is defined in } \mathbb{R}\}.$$

Strategy

Start from all real numbers and remove the values that violate a mathematical restriction.

Today the main restrictions will come from

denominators, even roots, logarithms, trigonometric denominators.

Interval notation: open and closed endpoints

Open interval

$$(a, b) = \{x : a < x < b\}$$



Closed interval

$$[a, b] = \{x : a \leq x \leq b\}$$



Round bracket (or) means the endpoint is excluded; square bracket [or] means it is included.

Intervals extending to infinity

Infinity is never included as an endpoint:

$$(-\infty, a), \quad (-\infty, a], \quad (a, +\infty), \quad [a, +\infty).$$



Unions of intervals are written with $\cup$, for example

$$(-\infty, -2) \cup (1, +\infty).$$

Functions defined for every real number

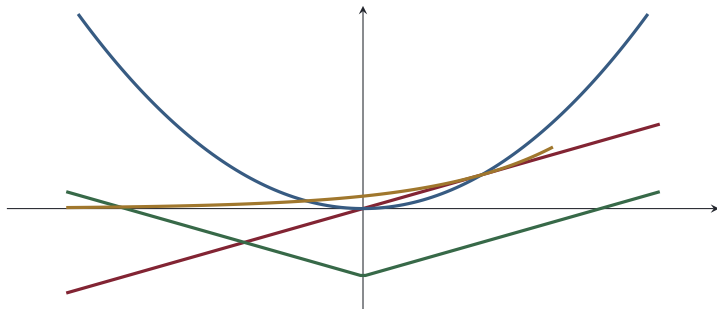
Some common families have natural domain $\mathbb{R}$.

Function type	Natural domain
Polynomial $P(x)$	$\mathbb{R}$
Absolute value $ g(x) $ when g is defined on $\mathbb{R}$	$\mathbb{R}$
Exponential a^x , $a > 0$	$\mathbb{R}$
Sine $\sin x$ and cosine $\cos x$	$\mathbb{R}$

Examples:

$$3x - 1, \quad x^4 - 2x^2 + 7, \quad |x - 3|, \quad e^x.$$

Different shapes, same domain



These graphs look very different, but each is defined for every real input.

$$D = \mathbb{R}.$$

Rational functions: denominator $\neq 0$

If

$$f(x) = \frac{P(x)}{Q(x)},$$

then we must require

$$Q(x) \neq 0.$$

Reason

Division by zero is not defined.

Example:

$$f(x) = \frac{x+1}{x-2} \implies x-2 \neq 0 \implies x \neq 2.$$

Hence

$$D = (-\infty, 2) \cup (2, +\infty).$$

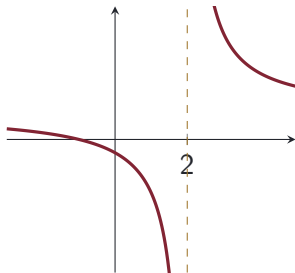
A missing input appears in the graph

For

$$f(x) = \frac{x+1}{x-2},$$

the input $x = 2$ is forbidden. The graph therefore has no point with horizontal coordinate 2.

$$D = \mathbb{R} \setminus \{2\}.$$



More than one forbidden denominator value

Exercise Find the domain of

$$f(x) = \frac{x+3}{x^2-4}.$$

Factor the denominator:

$$x^2 - 4 = (x-2)(x+2).$$

Thus

$$(x-2)(x+2) \neq 0 \implies x \neq 2, x \neq -2.$$

Therefore

$$D = \mathbb{R} \setminus \{-2, 2\}$$

or equivalently

$$D = (-\infty, -2) \cup (-2, 2) \cup (2, +\infty).$$

Simplifying a formula does not restore excluded points

Consider

$$f(x) = \frac{x-1}{(x-1)(x+3)}.$$

The original denominator requires

$$x \neq 1, \quad x \neq -3.$$

For admissible x ,

$$f(x) = \frac{1}{x+3}, \quad \text{but still} \quad \boxed{D = \mathbb{R} \setminus \{-3, 1\}}.$$

Important distinction

Simplifying the expression does not restore points excluded by the original formula.

Even roots: the radicand must be non-negative

For a real-valued square root,

$$\sqrt{g(x)} \text{ requires } g(x) \geq 0.$$

The same rule holds for every even root. Example:

$$f(x) = \sqrt{x - 2}$$

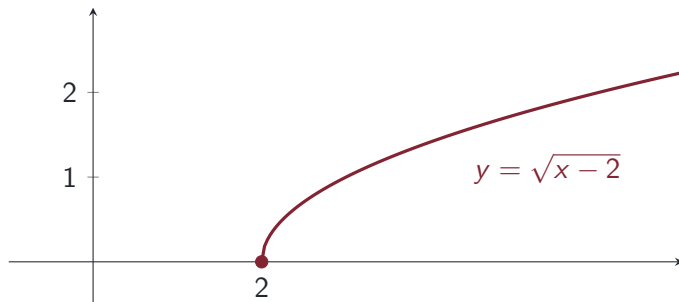
requires

$$x - 2 \geq 0 \implies x \geq 2.$$

Hence

$$D = [2, +\infty).$$

Why the graph starts at $x = 2$



There is no real value of $\sqrt{x - 2}$ when $x < 2$.

An even root can produce two separate intervals

Exercise Find the domain of

$$f(x) = \sqrt{x^2 - 4}.$$

We need

$$x^2 - 4 \geq 0 \iff (x - 2)(x + 2) \geq 0.$$

Sign analysis gives

$$x \leq -2 \quad \text{or} \quad x \geq 2.$$

Therefore

$$D = (-\infty, -2] \cup [2, +\infty).$$

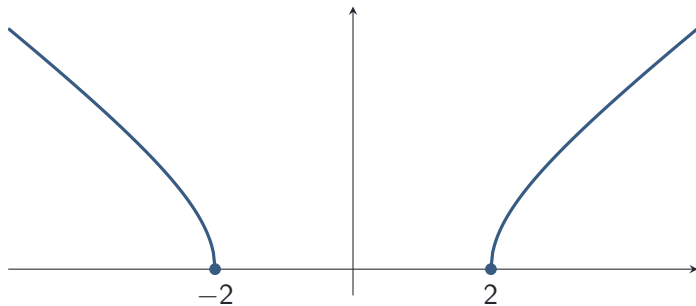
Sign chart for $x^2 - 4$



$$(x - 2)(x + 2) \geq 0$$

We keep the intervals where the expression is positive, and we include the zeros because the inequality is ≥ 0 .

The domain can have disconnected pieces



The function exists on two separate parts of the real line:

$$(-\infty, -2] \cup [2, +\infty).$$

Odd roots: no sign restriction

Odd roots are defined for every real radicand.

$$\sqrt[3]{g(x)}$$

does not require $g(x) \geq 0$. Example:

$$f(x) = \sqrt[3]{x-2}$$

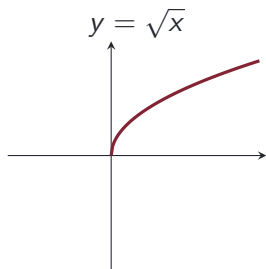
has

$$\boxed{D = \mathbb{R}.}$$

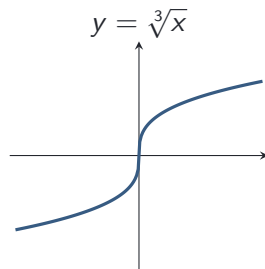
Negative values inside a cube root are perfectly admissible:

$$\sqrt[3]{-8} = -2.$$

Square root versus cube root



$$D = [0, +\infty)$$



$$D = \mathbb{R}$$

A square root in the denominator

Consider

$$f(x) = \frac{1}{\sqrt{x-2}}.$$

Two facts must hold simultaneously:

1. the square root must exist: $x - 2 \geq 0$;
2. the denominator must not vanish: $\sqrt{x-2} \neq 0$.

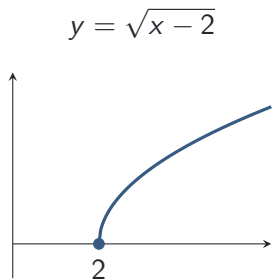
Together,

$$\boxed{x - 2 > 0} \implies \boxed{D = (2, +\infty)}.$$

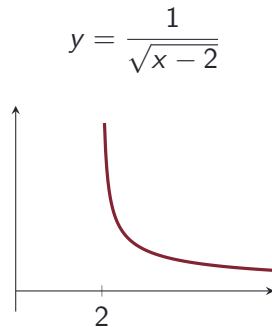
Rule

An even root in a denominator requires its radicand to be **strictly positive**.

Compare the two domains



$$D = [2, +\infty)$$



$$D = (2, +\infty)$$

Logarithms: the argument must be positive

For a real logarithm,

$$\log(g(x)) \text{ requires } g(x) > 0.$$

Zero is not allowed. Example:

$$f(x) = \ln(x - 3)$$

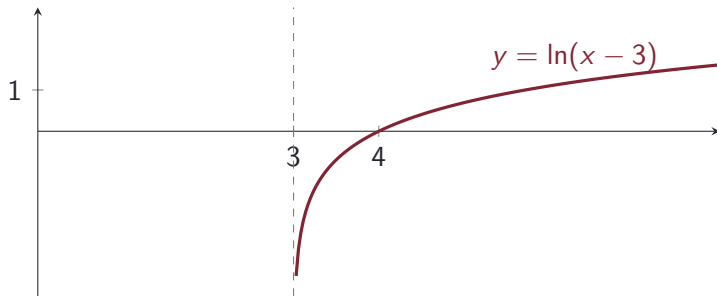
requires

$$x - 3 > 0 \implies x > 3.$$

Hence

$$D = (3, +\infty).$$

The graph of $\ln(x - 3)$ begins to the right of 3



The value $x = 3$ itself is excluded because $\ln 0$ is not defined.

Same internal expression, different inequality

Compare

$$\sqrt{x^2 - 4} \quad \text{and} \quad \ln(x^2 - 4).$$

For the square root:

$$x^2 - 4 \geq 0 \implies D = (-\infty, -2] \cup [2, +\infty).$$

For the logarithm:

$$x^2 - 4 > 0 \implies D = (-\infty, -2) \cup (2, +\infty).$$

Difference

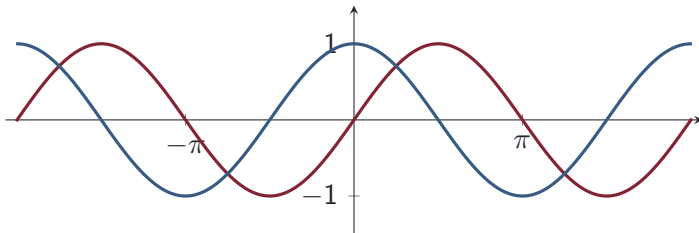
Square roots allow a zero radicand; logarithms do not allow a zero argument.

Trigonometric functions: sine and cosine

$\sin x,$ $\cos x$

are defined for every real number:

$$D = \mathbb{R}.$$



Tangent: denominator restrictions

Since

$$\tan x = \frac{\sin x}{\cos x},$$

we require

$$\cos x \neq 0.$$

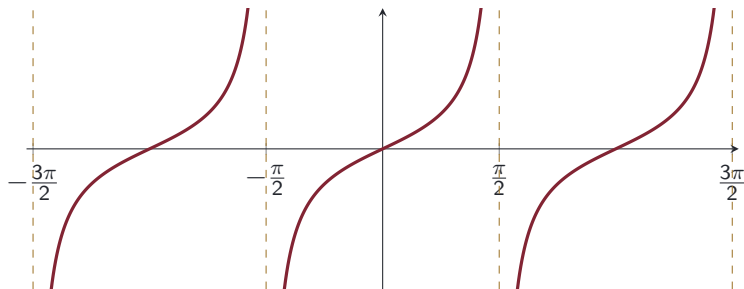
The forbidden values are

$$x = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

Therefore

$$D_{\tan} = \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi : k \in \mathbb{Z} \right\}.$$

The gaps in the tangent graph



The graph separates into branches because tangent is undefined whenever $\cos x = 0$.

Inverse trigonometric functions: useful domains

The inverse trigonometric functions have their own natural domains:

Function	Domain	Image
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

For example,

$$f(x) = \arcsin(2x)$$

requires

$$-1 \leq 2x \leq 1 \quad \Longrightarrow \quad \boxed{-\frac{1}{2} \leq x \leq \frac{1}{2}.}$$

Combining restrictions: intersection

Consider

$$f(x) = \frac{\sqrt{x-1}}{x-3}.$$

We need both

$$x - 1 \geq 0 \quad \text{and} \quad x - 3 \neq 0.$$

Thus

$$x \geq 1, \quad x \neq 3.$$

The domain is the **intersection** of the admissible sets:

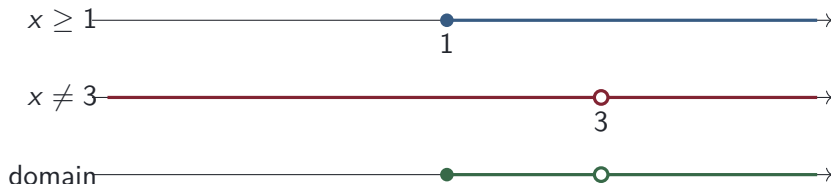
$$D = [1, 3) \cup (3, +\infty).$$

Visual intersection of restrictions

For

$$f(x) = \frac{\sqrt{x-1}}{x-3},$$

combine the two conditions.



A second combined example

Exercise Find the domain of

$$f(x) = \frac{\sqrt{x+2}}{x^2-1}.$$

Conditions:

$$x+2 \geq 0 \implies x \geq -2,$$

$$x^2-1 \neq 0 \implies x \neq -1, x \neq 1.$$

Therefore

$$D = [-2, -1) \cup (-1, 1) \cup (1, +\infty).$$

Logarithm of a rational expression

Exercise Find the domain of

$$f(x) = \ln \left(\frac{x-1}{x+2} \right).$$

The logarithm requires

$$\frac{x-1}{x+2} > 0.$$

Critical values:

$$x = -2, \quad x = 1.$$

The quotient is positive when numerator and denominator have the same sign.

Sign chart for the logarithm argument



Because the inequality is strict,

$$D = (-\infty, -2) \cup (1, +\infty).$$

The value -2 is forbidden by the denominator; the value 1 makes the logarithm argument equal to zero.

Nested functions: $\sqrt{\ln x}$

Consider

$$f(x) = \sqrt{\ln x}.$$

The outer square root requires

$$\ln x \geq 0.$$

This condition automatically implies $x > 0$, so the logarithm also exists. Since

$$\ln x \geq 0 \iff x \geq 1,$$

we obtain

$$D = [1, +\infty).$$

Nested functions: $\ln(\sqrt{x-2})$

Consider

$$f(x) = \ln(\sqrt{x-2}).$$

The logarithm requires its argument to be strictly positive:

$$\sqrt{x-2} > 0.$$

Therefore

$$x - 2 > 0 \implies \boxed{D = (2, +\infty)}.$$

Notice

Although $\sqrt{x-2}$ alone allows $x = 2$, placing it inside a logarithm excludes that endpoint.

Domain of a composition

For a composition

$$(g \circ f)(x) = g(f(x)),$$

we need

$$x \in D_f \quad \text{and} \quad f(x) \in D_g.$$

Example:

$$f(x) = x^2 - 4, \quad g(x) = \sqrt{x}.$$

Then

$$(g \circ f)(x) = \sqrt{x^2 - 4}$$

and therefore

$$D = (-\infty, -2] \cup [2, +\infty).$$

The order of composition changes the domain

Using

$$f(x) = x^2 - 4, \quad g(x) = \sqrt{x},$$

we found

$$(g \circ f)(x) = \sqrt{x^2 - 4}, \quad D_{g \circ f} = (-\infty, -2] \cup [2, +\infty).$$

But

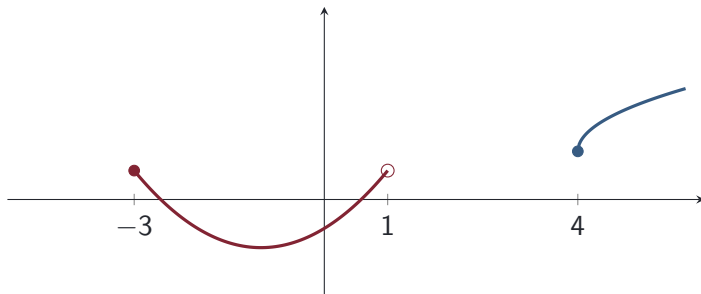
$$(f \circ g)(x) = (\sqrt{x})^2 - 4 = x - 4$$

is only defined where $\sqrt{x}$ exists:

$$D_{f \circ g} = [0, +\infty).$$

The simplified formula $x - 4$ does not erase the original restriction $x \geq 0$.

Reading the domain directly from a graph



The graph exists for

$$D = [-3, 1) \cup [4, +\infty).$$

Closed dot: endpoint included. Open dot: endpoint excluded.

Domain algorithm

For a real-valued function, use this checklist.

1. Start from $\mathbb{R}$.
2. Exclude zeros of denominators.
3. Require even-root radicands to be ≥ 0 .
4. If an even root is in a denominator, require its radicand to be > 0 .
5. Require logarithm arguments to be > 0 .
6. Apply trigonometric or inverse-trigonometric restrictions.
7. Intersect all conditions.
8. Write the final set in interval notation.

Habit to develop

Write the restrictions first; simplify only afterwards.

Guided practice I

Exercise Determine the domain of each function.

$$A: \frac{1}{x+3}, \quad B: \sqrt{2x-4}, \quad C: \ln(5-x).$$

Solution

$$D_A = \mathbb{R} \setminus \{-3\},$$

$$2x - 4 \geq 0 \Rightarrow x \geq 2 \quad \Rightarrow \quad D_B = [2, +\infty),$$

$$5 - x > 0 \Rightarrow x < 5 \quad \Rightarrow \quad D_C = (-\infty, 5).$$

Guided practice II

Exercise Determine the domain of

$$f(x) = \sqrt{x^2 - 9}.$$

Solution We need

$$x^2 - 9 \geq 0 \iff (x - 3)(x + 3) \geq 0.$$

Thus

$$x \leq -3 \quad \text{or} \quad x \geq 3,$$

so

$$D = (-\infty, -3] \cup [3, +\infty).$$

Guided practice III

Exercise Determine the domain of

$$f(x) = \frac{\ln(x-1)}{\sqrt{5-x}}.$$

Conditions:

$$x - 1 > 0 \quad \implies \quad x > 1,$$

$$5 - x > 0 \quad \implies \quad x < 5,$$

because the square root is in the denominator. Therefore

$$D = (1, 5).$$

Guided practice IV: a rational expression under a root

Exercise Determine the domain of

$$f(x) = \sqrt{\frac{x-2}{x+1}}.$$

We need

$$\frac{x-2}{x+1} \geq 0, \quad x \neq -1.$$

Critical points: -1 and 2 . The quotient is non-negative on

$$(-\infty, -1) \cup [2, +\infty).$$

Hence

$$D = (-\infty, -1) \cup [2, +\infty).$$

Guided practice V: logarithm and quotient

Exercise Determine the domain of

$$f(x) = \ln \left(\frac{x+3}{x-1} \right).$$

We need

$$\frac{x+3}{x-1} > 0.$$

Critical values are -3 and 1 . The quotient is positive when

$$x < -3 \quad \text{or} \quad x > 1.$$

Therefore

$$D = (-\infty, -3) \cup (1, +\infty).$$

Guided practice VI: several restrictions at once

Exercise Determine the domain of

$$f(x) = \frac{\sqrt{x-1}}{(x-3)\ln(x+2)}.$$

Restrictions:

$$x - 1 \geq 0 \Rightarrow x \geq 1,$$

$$x \neq 3,$$

$$x + 2 > 0,$$

and because $\ln(x+2)$ is in the denominator,

$$\ln(x+2) \neq 0 \Rightarrow x+2 \neq 1 \Rightarrow x \neq -1.$$

After intersecting all conditions with $x \geq 1$, only $x = 3$ must still be removed. Hence

$$D = [1, 3) \cup (3, +\infty).$$

Independent practice

Exercise Determine the natural domain of each function.

1. $f(x) = \frac{x+1}{x^2-5x+6}$

2. $g(x) = \sqrt{4-x^2}$

3. $h(x) = \frac{1}{\sqrt{x^2-4}}$

4. $p(x) = \ln(x^2-x-6)$

5. $q(x) = \sqrt{\ln(x-1)}$

6. $r(x) = \ln(\sqrt{6-x})$

Independent practice: answers

$$D_f = \mathbb{R} \setminus \{2, 3\},$$

$$D_g = [-2, 2],$$

$$D_h = (-\infty, -2) \cup (2, +\infty),$$

$$D_p = (-\infty, -2) \cup (3, +\infty),$$

$$D_q = [2, +\infty),$$

$$D_r = (-\infty, 6).$$

Check the endpoint logic

A square root may allow equality; a logarithm and a denominator may remove that endpoint again.

What you should be able to do after this lecture

- ▶ Translate inequalities into interval notation.
- ▶ Find forbidden denominator values.
- ▶ Distinguish even-root and odd-root restrictions.
- ▶ Distinguish ≥ 0 for square roots from > 0 for logarithms.
- ▶ Handle roots and logarithms in denominators.
- ▶ Combine several restrictions by intersection.
- ▶ Determine domains of nested functions and compositions.
- ▶ Read a domain directly from a graph.

Looking ahead: from existence to sign

Today we asked:

Where does $f(x)$ exist?

The next question is:

Where is $f(x) > 0$, $f(x) = 0$, $f(x) < 0$?

Lecture 12 will develop

- ▶ zeros and intercepts;
- ▶ sign analysis;
- ▶ sign charts;
- ▶ intersections between graphs.